

Section - A

A1

- i. Landmarks for localization
- ii
 1. LiDAR and wheel encoders
 2. A mobile robot uses wheel encoders and LiDAR for localization, over time, errors in sensor readings can accumulate, affecting navigation accuracy and path planning.

A2.

- i. a. Localization becomes less accurate due to loss of stable reference points.
- ii
 - a. True
Errors in sensor readings can accumulate affecting navigation accuracy and path planning.
 - b. Errors in sensor reading can accumulate, affecting navigation accuracy and path planning. Mobile robot uses sensors for localization, errors in measurement / sensor readings affect the navigation accuracy

Section B.

B1.

i. C Ignoring environmental effects

ii. d ops localization

iii

a

b

e

d

f

c

iv. a. Noise

while passing a glass walls and shiny surface the sonar beams bounce irregularly and produce false readings

b. Aliasing

while robot enters two identical looking corridors, its sensor data cannot distinguish the two and get confused of two similar corridors.

B3.

- i. b It builds and updates maps while localizing
- ii. b Sensor fusion for localization
- iii. Belief representation helps drone to improve localization by using inputs like wall distance and camera readings.
 - * There is no need of GPS
 - * Instead of assuming a single fixed position, it maintains all possible locations.
 - * It increases certainty.
- iv.
 1. Mapping combined with planning
 2. Navigation becomes guesswork.
 3. Continuous updates prevent collisions
 4. Dynamic re-routing is supported
 5. Obstacle detection is unnecessary.

Section - C

C2.

- i. c. Distributed system with shared environment models.

ii. a use sensor fusion techniques to combine reading intelligently.

iii. $q = (2, 2)$

$q_{obs} = (4, 4)$

$k_{rep} = 100$

$p_0 = 3m$

$$P(q) = \sqrt{(2-4)^2 + (2-4)^2}$$

$$= \sqrt{4+4}$$

$$= \sqrt{8}$$

$P(q) = 2.82$

$P(0) > P(q)$

$3 > 2.82$

iv. In order to make warehouse robot more adaptable it should be trained to adapt the dynamic conditions like newly placed shelf. As it follows the pre-planned path, when it encounters a new things it will be hard to identify so it should be designed by the way that should get adapted for dynamic changes.

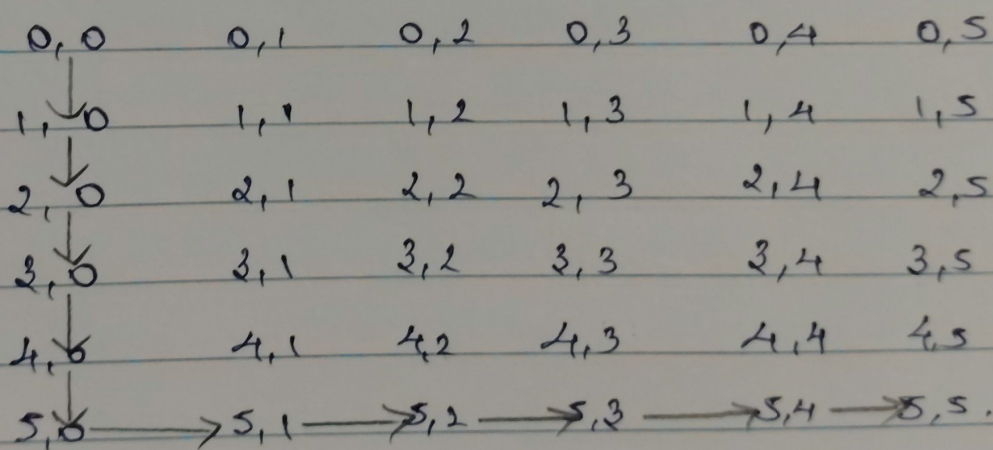
c3.

- i. d. Recompute the path based on updated map
- ii. c Take an alternative route using stairs or another elevator.

iii. DWA - Dynamic window Approach

DWA is best categorized as a real time, reactive, local path planning and obstacle avoidance techniques, critical for safe and adaptive movement in unpredictable and hazardous environments.

iv. optimal path



0 - free cell

1 - Road or wall

Path - 0,0 ; 1,0 ; 2,0 ; 3,0 ; 4,0 ; 5,0 ; 5,1 ;
5,2 ; 5,3 ; 5,4 ; 5,5.